Third Research of Astra 6 Max

Round 3 | Deep third cumulants in random ReLU networks

I cannot demonstrate a method meeting both the requested cost and 2% accuracy. The four suggested reformulations retain the expensive mixed-index information. One different, untested approximation does have a concrete sub-1.2-u implementation: approximate the **birth matrix on the right** by diagonal plus small rank. Its accuracy is the unresolved question.

Derived = algebra or cost accounting; Conjecture = unverified interpretation; Measure-this = proposed experiment. Network measurements below are yours. I have not run the proposed test on your checkpoints. Costs use float32 and u = 2^{31} FLOPs; small rounding allowances are explicit.

1. Sharing squared propagators

Derived. Write T_b=CX. The exact identity is

$$(CX)^{\circ 2} = C^{\circ 2}X^{\circ 2} + 2\sum_{p<q}(C_{:p}\circ C_{:q})(X_{p:}\circ X_{q:}).$$

The second term contains the cross-column products. Squaring is not a matrix homomorphism. For example, rows (1,1) and (1,−1) have identical elementwise squares, but multiplication by $x=(1,1)^T$ followed by squaring gives 4 and 0. Thus C∘C and X alone cannot determine the result.

An exact lifted state keeps all symmetric pairs, dimension d=n(n+1)/2. Its update is the symmetric-square representation of matrix multiplication. An n×d evaluation array alone occupies **2.002 GiB**; a d×n coefficient array is equally large *per birth*. Directly multiplying them costs

$$\frac{n^2(2d-1)}{2^{31}} = 512.50 \text{ u}.$$

At inner rank r=384, d=73,920 and that contraction still costs **72.19 u**. This is n^2r^2-class work. Contracting the birth index first instead produces the pooled cubic core already measured in item 7.

Keeping the cross terms preserves the realized fluctuation; discarding them supplies no controlled approximation. The 2×2 example falsifies an exact squares-only closure essentially for free. Your diagonal-chaos and pooled-core tests already address the obvious approximate versions; I would not repeat them.

The tensor-power composition law is standard; see van Neerven, section 3. It establishes composition, not a cheaper contraction algorithm.

2. Wick expansion: low degree, many coefficients

Derived. Condition on earlier layers. Put V=D $T_{previous}$, U=D $\tilde{A}_{previous}$, and $W_{i:}=\sigma g_i^\top$, $\sigma^2=2/n$. For columns v_r, u_r,

$$F_{ir} = \sigma^2 \langle g_i g_i^\top - I,\ v_r v_r^\top \rangle,$$

$$(F\widetilde{A}^\top)_{ik} = \sigma^3 \sum_{abc} (g_{ia} g_{ib} - \delta_{ab}) g_{kc}\, \mathcal{C}_{abc},$$

$$\mathcal{C}_{abc} = \sum_r v_{ar} v_{br} u_{cr}.$$

Here F subtracts the conditional Gaussian row mean; other row-constant centerings differ by a thin term. This is exact quadratic chaos times a linear factor. The coefficients are precisely the cubic tensor we were trying to avoid. For i=k, further Wick decomposition separates cubic and linear terms; it does not eliminate the cubic coefficients. Averaging the expression destroys its realized odd value.

The aggregate state updates as $K^+=M^{\otimes 3}K+K_{birth}$. A symmetric rank-r core has r(r+1)(r+2)/6 scalars. Using even your favorable core costs at the measured r≈250:

$$c_{\text{layer}} \geq \frac{nr^3/2 + nr^3/2 + 3r^4}{2^{31}} = 12.91 \text{ u}.$$

Those are slice, insertion and rotation costs, before other work. At r=256 the sum is 14 u. The state is only 2.64 million coefficients at r=250, but updating and querying it is expensive. Item 7 has already falsified the affordable rank reduction; another random-feature test is unwarranted.

3. Backward response: the same tensor transport

Derived. After fixing the Gaussian-chain linearization, one scalar projection of the final mean has adjoint recurrence

$$\Lambda_l = (M_{l+1}^\top)^{\otimes 3} \Lambda_{l+1} + Q_l, \qquad \delta f = \sum_b \langle \Lambda_b, K_b \rangle.$$

Q_l is supported on the queried slices. Backward mixing immediately creates off-slice entries. A factored adjoint source again has two dense legs: two transport products and two slice products when pairing with a birth, approximately **2.05 u per query-source/layer**. This is an implementation count, not a universal lower bound.

A forward/backward split replaces d transport steps by h+(d−h), and adds a joining contraction. Moreover, one scalar adjoint does not produce all 1024 means. A direct block of 1024 rank-250 symmetric cores would exceed 10 GB. No cheaper complete update follows from reversing the arrows.

A short backward horizon simply discards long response paths unless another state carries them. Your sensitivity measurements already reject the straightforward truncation. They do not prove that every possible response compression is impossible.

4. Memory closures and contraction order

Derived. Split the tensor into resolved slices q and hidden entries z. The exact linear stage has the block form

$$q^+ = A_{qq}q + A_{qz}z + s_q, \qquad z^+ = A_{zq}q + A_{zz}z + s_z.$$

Eliminating z gives a memory kernel plus a realized forcing term. Keeping that term retains the odd information; replacing it by its ensemble mean loses it. A generic kernel between n^2-dimensional slices has n^4 entries: **4 TiB** in float32 and approximately **1024 u per lag application**. A factorization must supply an actual cheaper realization, which the formal identity does not provide.

This is the relevant lesson of Mori-Zwanzig, rather than a new closure to try. The memory-cost issue is explicit in Li and Stinis (2019). Tensor-network contraction of the birth index first returns the same cubic core. Your measured rank and slice-prediction failures reject the specified inexpensive versions, not every conceivable memory model.

5. One conditional alternative: compress the birth operator

Conjecture to test. Approximate each fixed birth matrix by

$$A_b \approx \widehat{A}_b = \mathrm{diag}(d_b) + U_b V_b^\top, \qquad U_b, V_b \in \mathbb{R}^{n \times k}.$$

This keeps **all n hubs and every birth**. It does not truncate the left propagator space or the Hadamard products. Full-rank A_b does not itself exclude this representation because its diagonal remains full rank. None of the listed spectra measures this particular approximation.

Derived - state and update. Per birth retain T, d, U, V: $n^2+2nk+n$ floats, **4.254 MiB at k=32**, plus reusable workspace. Update only T←MT. Define

$$R = TU, \qquad E = RV^\top, \qquad S = T \circ T.$$

Substituting $\tilde{A}=T\,\mathrm{diag}(d)+E$ in the exact slice formula gives

$$\widehat{D}_{21} = \left[S\,\mathrm{diag}(d) + \tfrac{2}{3}(T \circ E)\right]T^\top + \tfrac{1}{3}(SV)R^\top.$$

The diagonal birth contribution merges the two dense slice products. There is no separately transported dense A leg. Crucially, S is computed exactly and may remain full rank. Under the latest W→−W, T,R,E change sign while S does not; the entire readout changes sign. This preserves a realized odd fluctuation, rather than predicting its zero conditional mean.

5, continued. Complete arithmetic and fitting

Derived - per source-layer cost. There are two dense products: MT and the bracketed expression times T^T. The other four are TU, RV^T, SV and $(SV)R^T$. For $k \geq 1$, their exact total is $8n^2k - 2nk - 2n^2$ FLOPs. Including an allowance for elementwise operations,

$$c(k) \leq 2s + \frac{4k}{n} + 0.005 \text{ u},$$

where s is the metered Strassen product cost. Conservatively taking s=2.05/4 gives **1.030, 1.093, 1.155 and 1.186 u** for k=0,16,32,40. Thus $k \leq 40$ meets the update target. This count includes transport and readout; fitting is priced next.

At k=32, all 120 source-layers cost **138.6 u**. With the fit below, the cumulant bill is at most **143.1 u, or 1.193 u per source-layer amortized**. Add C_{other} for the rest of the moment chain; its complete bill is not certified. Also, the two requested targets are not pointwise equivalent: ten dense old sources here cost 11.55 u per layer. This is a conditional route to the first target only.

Measure-this - specified fit. Initialize d=diag(A). Compute one randomized rank-k SVD of $A - \text{diag}(d)$, then reset $d = \text{diag}(A - UV^T)$. Use oversampling 16 and one power iteration. The final reset preserves the exact birth diagonal.

With $q = k + 16 \leq 56$, the fit uses four $n \times n$ by $n \times q$-type products, about $4q/n \leq 0.219$ u, plus QR and the small SVD. Allow **0.3 u per birth, 4.5 u for 15 births**. The fitted d,U,V stay fixed during transport. This is a reproducible candidate fit, not the globally optimal fit for the later query norm. At k=40, fitting raises the amortized bill to 1.224 u; use k=32 for the strict target.

Conjecture - likely outcome. I would expect failure if the off-diagonal birth matrix really behaves like isotropic noise. Entries of size $n^{-1/2}$ across n^2 locations have $O(\sqrt{n})$ Frobenius norm, comparable to an O(1) diagonal. They are not collectively small. Nevertheless, their transported *query* norm, rather than their entrywise scale, decides this test.

The approximation error is exactly the linear image of $\Delta A = A - \text{diag}(d) - UV^T$ under the given transport/readout. It can accumulate coherently across births. Neither parity nor the exact algebra supplies a 2% accuracy guarantee.

6. First falsifier and what the leaders imply

Measure-this. On one network, cache the birth matrices and V29 legs at layers 6, 10 and 14. Fit k=32. At each checkpoint substitute the fitted birth matrices, leaving the cached T fixed, and compare **the sum over all old births** with the exact sum. Report total D21, its diagonal, and separately the fluctuation contribution

$$\sum_b F_b \widetilde{A}_b^\top, \qquad F_b = T_b^{\circ 2} - \mathbf{1} m_b^\top.$$

Use the same exact m_b in both evaluations. Also inspect the mixed slice term separately. A large trace component must not conceal an inaccurate fluctuation. Require 2% relative Frobenius error for the stipulated fluctuation and D21 queries at every checkpoint. Inspect per-birth errors to diagnose cancellation; the aggregate is the primary acceptance test.

From cached legs, allow **under 100 u** for fitting and these checks. Failure rejects this specified fit immediately, not every possible query-weighted diagonal-plus-rank fit. Only if it passes, rerun the whole chain with gates and new birth matrices recomputed; check every layer's full D21, compare against your existing truth, and meter the complete bill.

A baseline plus the checkpoint test and a conditional full rerun can be capped near **650 u**: 1.40×10^{12} metered FLOPs, about 23 minutes at sustained 1 GFLOP/s. The first cached-checkpoint falsifier needs about four minutes at that throughput; actual CPU time should be calibrated on one matrix product.

Derived - accuracy budget. Your error law assigns ε=0.02 an extra MSE of 1.68×10^{-9}, about 8% of V29's error. Under that additive law, allowing 1.3× V29's MSE corresponds to ε≈3.9%. Thus a successful right-operator compression could lose less than 1.3× without dropping births. The additive error law alone does not explain the leaders' *improved* raw accuracy; that would require a demonstrated reduction in another error.

Conjecture, with limited evidence. If the leaders use this kind of algebraic simplification, the discriminating observable is that their birth operators admit k≲32 query-accurate diagonal-plus-rank fits, and their bill scales like two dense products plus thin corrections. The current measurements neither establish that signature nor identify another leader mechanism. Your new item 10 specifically kills my previous coupled-closure retuning hypothesis.

Derived - the extra accuracy rung is not identified. Similar MSE ratios at smaller widths do not prove that leaders propagate K4 or all augmented terms. A full symmetric fourth cumulant at n=1024 already contains 46,081,900,800 floats, **171.7 GiB**, before workspace. I have no cost-complete cheap higher-order update to offer; attributing their gain to one would be speculation.

I cannot justify a universal 2-u floor. 2.05 u is the cost of the known four-product formulation, not a proved arithmetic floor. The strongest obstruction is that all measured cheap states lose the realized mixed-index information, while the measured accurate pooled state costs over 12 u per insertion layer. If the right-operator test also fails, I cannot name an approximation that halves the bill within the stated accuracy loss, and I would not infer one from the leaderboard numbers alone.